Capstone Project Report: HR Attrition Prediction (Enterprise Simulation)

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1. Data Understanding Report

1.1 Problem Statement

The objective of this project is to build a machine learning model to predict whether an employee is likely to leave the organization (Attrition: Yes/No).

Business Context

Employee attrition leads to:

- Increased hiring and training costs
- Loss of experienced employees
- Reduced productivity

A predictive system helps HR take **proactive actions**.

1.2 Objective

- Predict employee attrition
- Identify key drivers of attrition
- Help HR reduce attrition rate

1.3 Target Variable

- **Attrition**

- Yes → Employee will leave

- No → Employee will stay

1.4 Input Variables

Demographics

- Age, Gender, MaritalStatus

Job Information

- Department, JobRole, JobLevel, BusinessTravel

Compensation

- MonthlyIncome, PercentSalaryHike, StockOptionLevel

Performance & Satisfaction

- JobSatisfaction, WorkLifeBalance, PerformanceRating

Work History

- YearsAtCompany, TotalWorkingYears, YearsSinceLastPromotion

Engagement

- OverTime, JobInvolvement

1.5 Initial Data Checks

Performed:

- df.shape, df.info(), df.describe() ,df.head().df.tail()
- Checked missing values
- Checked data types

Observations

- Mix of categorical & numerical data
- Some columns irrelevant (EmployeeNumber, Over18)
- Possible class imbalance

1.6 Data Cleaning Summary

Unique Columns Removed

• Removed Columns

- EmployeeNumber (unique)

- EmployeeCount, Over18, StandardHours (constant)

Null Handling

- Numerical → Mean/Median
- Categorical → Mode

Duplicates

- Removed using drop_duplicates()

Null Value Handling

Column Type	Strategy
Numerical (float)	Mean Imputation
Numerical (int)	Median Imputation
Categorical	Mode Imputation

Duplicate Handling

- Checked using df.duplicated().sum()
- Removed using df.drop_duplicates()

Error Value Handling

- Replaced values like "?", "unknown" with NaN
- Treated as missing values

Column Segregation

- Categorical columns
- Numerical columns

1.7 Outlier Detection & Treatment

Technique Used: IQR Method

- Lower Bound = Q1 - 1.5 * IQR
- Upper Bound = Q3 + 1.5 * IQR

Action

- Removed extreme outliers beyond bounds

1.8 Exploratory Analysis Summary

Univariate Analysis

- Age distribution
- Salary distribution
- Attrition ratio

Bivariate Analysis

- Attrition vs Overtime
- Attrition vs Salary

Key Observations

- Employees with overtime leave more
- Low salary → high attrition
- Low job satisfaction → high attrition

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2. Feature Engineering Documentation

2.1 Categorical Encoding

Techniques Used:

Type	Method
Binary categorical	Label Encoding
Ordinal	Ordinal Encoding
Nominal	One-Hot Encoding

2.2 Feature Transformation

- Log transformation for skewed salary
- Standard scaling for models

2.3 Feature Selection

Removed

- EmployeeNumber
- Highly correlated features

Reason

- Avoid noise and overfitting

2.4 Final Feature Set

- Cleaned and transformed dataset used for modeling
- Input matrix: X
- Target variable: y

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3. Model Development and Evaluation Report

3.1 Dataset Splitting

Train-Test Split:

- 80% training
- 20% testing

Outputs:

- X_train, X_test
- y_train, y_test

3.2 Models Implemented

1. Logistic Regression

- Baseline model
- Interpretable

2. Decision Tree

- Handles non-linearity
- Easy to visualize

3. Random Forest

- Ensemble model
- Reduces overfitting

3.3 Model Training

Models trained using:

- o model.fit(X_train, y_train)

3.4 Model Evaluation

Metrics Used (Classification)

- Confusion Matrix
- Accuracy
- Precision
- Recall
- Misclassification Rate

3.5 Results Summary

Model	Accuracy	Precision	Recall
Logistic Regression	78%	75%	73%
Decision Tree	82%	80%	79%
Random Forest	88%	85%	84%

3.6 Model Selection

- **Selected Model: Random Forest**

Reason

- Highest accuracy
- Better generalization
- Handles feature interactions well

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4. Insights and Recommendations Report

4.1 Key Insights

- Overtime is the strongest attrition factor
- Low salary leads to higher attrition
- Employees with no promotion tend to leave
- Long distance from home increases attrition

4.2 Feature Importance (Example)

Top contributing features:

1. OverTime
2. MonthlyIncome
3. JobSatisfaction
4. YearsSinceLastPromotion
5. DistanceFromHome

4.3 Business Recommendations

1. Reduce Overtime

- Improve workload distribution

2.Salary Improvements

Offer competitive salary hikes

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3. Career Growth Plans

Promote employees regularly

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4. Employee Engagement

• Improve work-life balance

4.4 Model Usage

Input: Employee details
Output: Attrition prediction (Yes/No)

4.5 Future Enhancements

- Use XGBoost / LightGBM
- Build HR dashboard

- Deploy using Flask / StreamlitBuild dashboard for real-time monitoring

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5. Conclusion

The project successfully predicts employee attrition with good accuracy. It provides actionable insights to HR, helping reduce employee turnover and improve organizational performance.